

ASSIGNMENT – 1

Submitted by:

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25MCA044

Semester:2

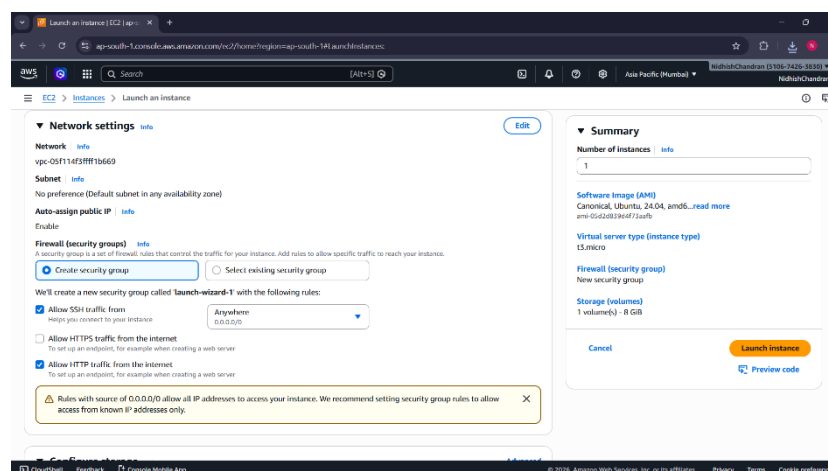
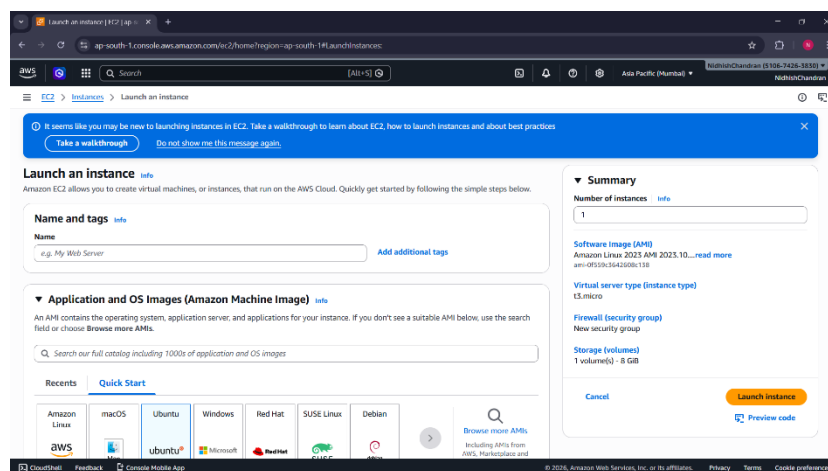
Hosting a Custom Website on Ubuntu EC2 Instance using Apache Web Server

Objective: To host a custom website by installing Apache web server on an Ubuntu EC2 instance and make it accessible globally using its public IPv4 address.

Procedures

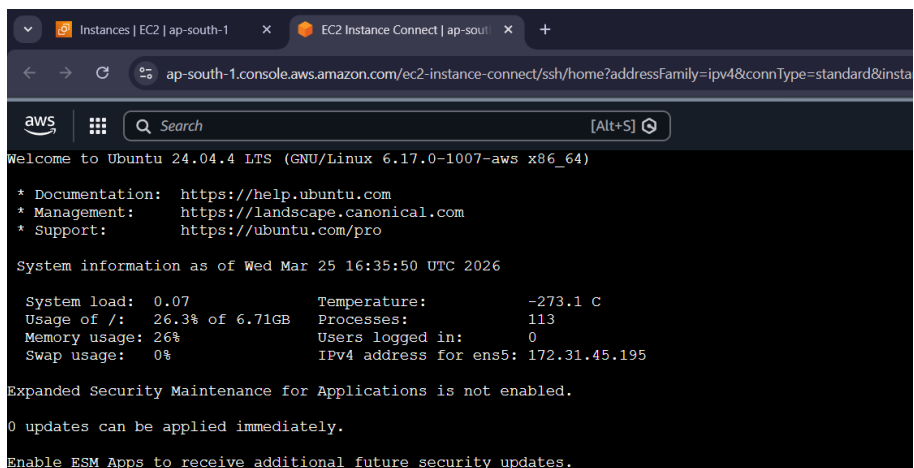
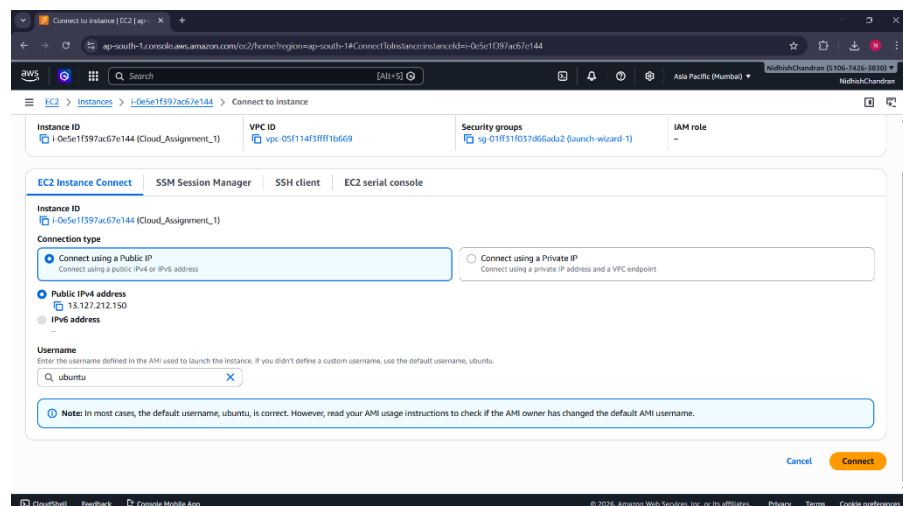
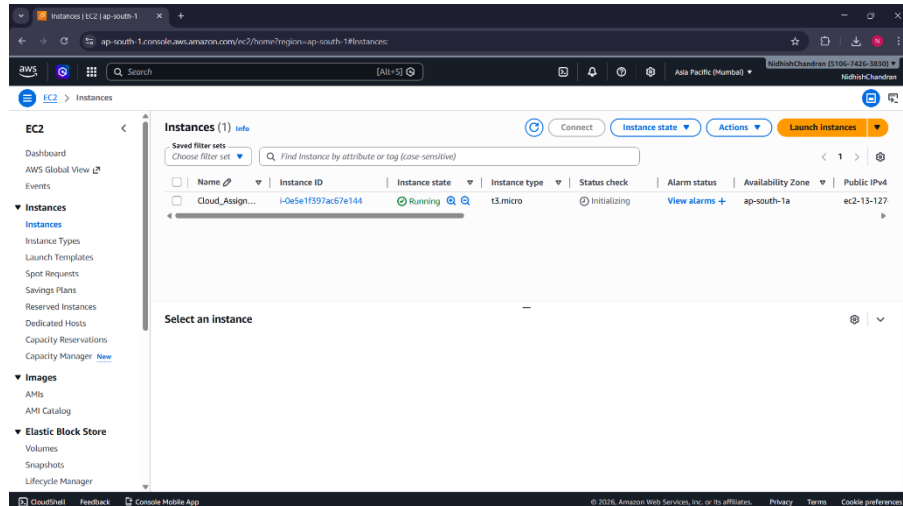
Step 1: Launch EC2 Instance

- Login to AWS Console
- Go to EC2 → Launch Instance
- Select Ubuntu Server (Latest)
- Choose t2.micro
- Create key pair
- Configure Security Group:
 - Allow SSH (Port 22)
 - Allow HTTP (Port 80)
- Launch instance



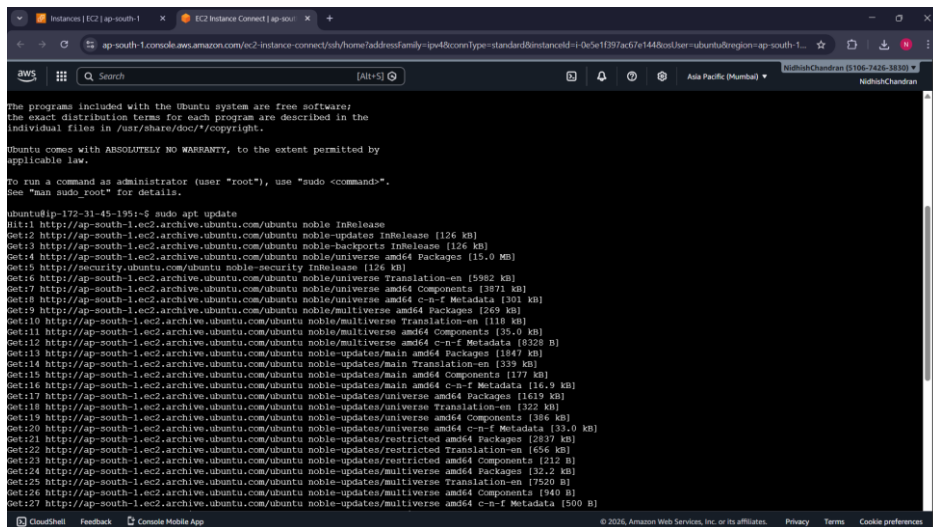
Step 2: Connect to EC2 Instance

- Go to EC2 → Instances
- Click Connect → EC2 Instance Connect → Connect



Step 3: Update System

- `sudo apt update`



```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

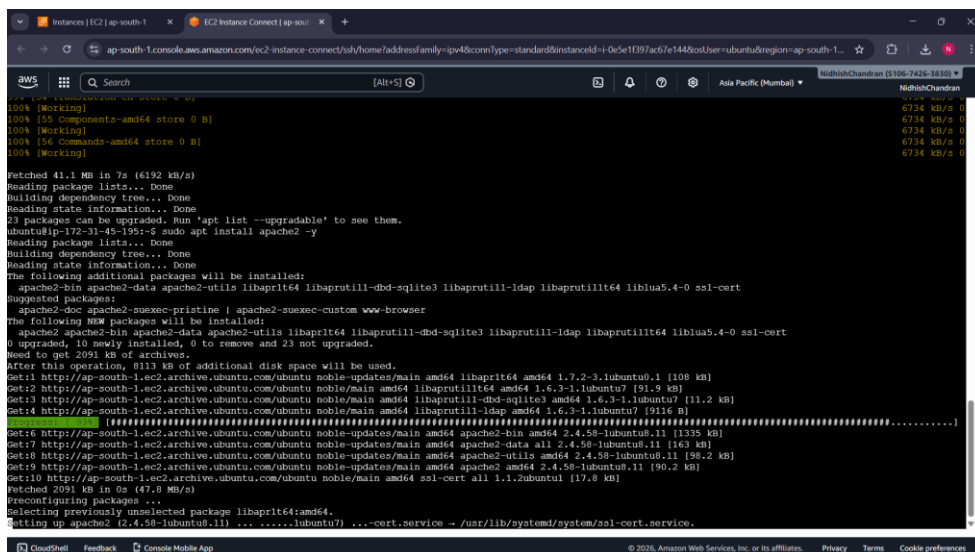
ubuntu@ip-172-31-45-195:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1847 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1619 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [322 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [186 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2837 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [656 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.2 kB]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [949 B]
Get:27 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]
```

Step 4: Install Apache Web Server

- `sudo apt install apache2 -y`

Start Apache:

- `sudo systemctl start apache2`

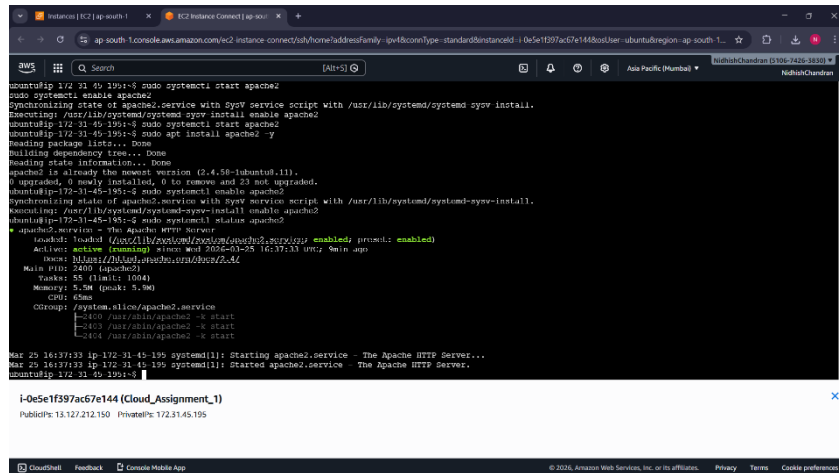


```
100% [Working]
100% [55 Components-amd64 store 0 B]
100% [Working]
100% [56 Commands-amd64 store 0 B]
100% [Working]

patched 41.1 MB in 7s (6192 KB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
23 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-45-195:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
  apache2-doc apache2-ssl-session-cache apache2-ssl-session-cache-browser
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 23 not upgraded.
Need to get 2091 kB of archives.
After this operation, 8113 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [100 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.11 [1335 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.11 [163 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.11 [98.2 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.11 [90.2 kB]
Patched 2091 kB in 0s (47.8 MB/s)
Preconfiguring packages
Selecting previously unselected package libapr1t64:amd64.
Setting up apache2 (2.4.58-1ubuntu8.11) ...
.....libubuntu7) ....cert.service - /usr/lib/systemd/system/ssl-cert.service.
```

Step 5: Verify Apache Status

- `sudo systemctl status apache2`



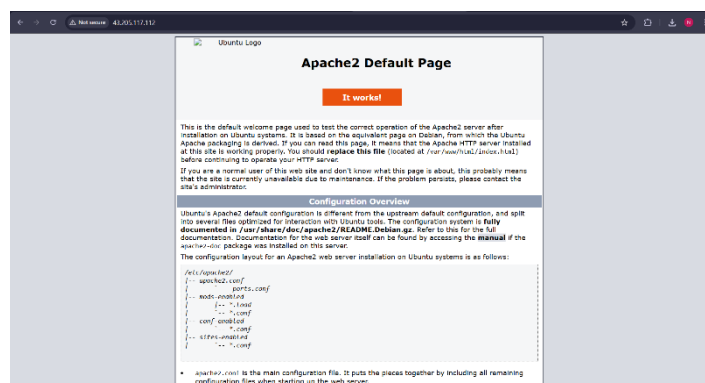
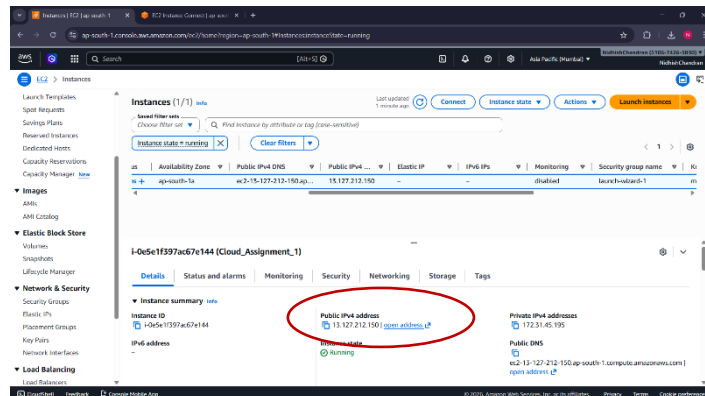
```
mountu8ip-172-31-45-195~$ sudo systemctl start apache2
sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
mountu8ip-172-31-45-195~$ sudo systemctl start apache2
mountu8ip-172-31-45-195~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.56-1ubuntu1.1).
0 upgraded, 0 newly installed, 0 to remove and 23 not upgraded.
mountu8ip-172-31-45-195~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
mountu8ip-172-31-45-195~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2024-03-28 16:37:33 UTC; 9min ago
     Main PID: 2400 (apache2)
    CGroup: /system.slice/apache2.service
            └─2400 /usr/sbin/apache2 -k start
            └─2403 /usr/sbin/apache2 -k start
            └─2404 /usr/sbin/apache2 -k start

Mar 28 16:37:33 ip-172-31-45-195 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 28 16:37:33 ip-172-31-45-195 systemd[1]: Started apache2.service - The Apache HTTP Server.
mountu8ip-172-31-45-195~$
```

I-0e5e1f97ac67e144 (Cloud_Assignment_1)
PublicIPs: 13.127.212.150 PrivateIPs: 172.31.45.195

Step 6: Access Default Web Page

- Copy Public IPv4 address
- Open browser



Step 7: Create Custom Website

Open file:

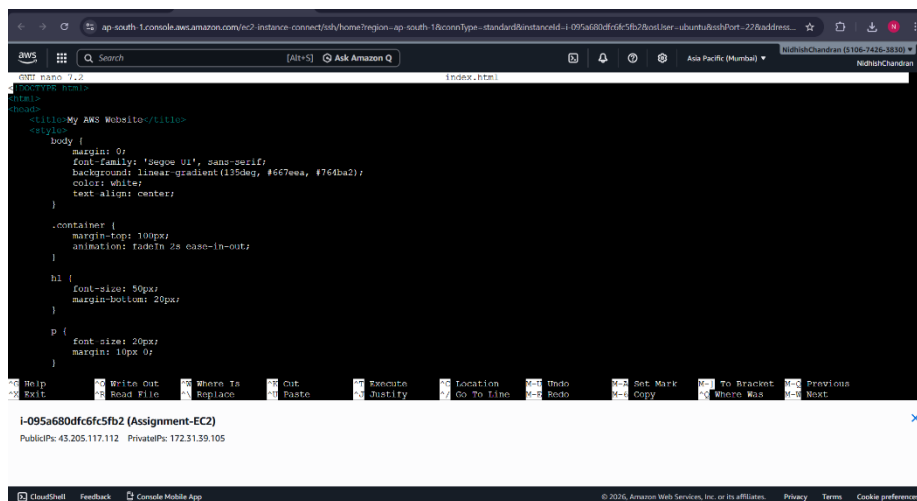
- `cd /var/www/html`
- `sudo nano index.html`

Add Html:

- `<h1>Welcome to My AWS Website</h1>`
- `<p>This website is hosted using EC2. </p>`

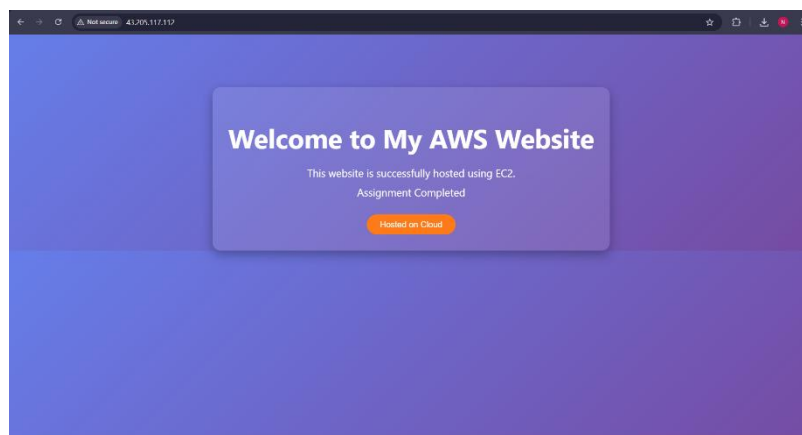
Save and exit

```
Mar 25 17:12:33 ip-172-31-39-105 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Mar 25 17:12:33 ip-172-31-39-105 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-39-105:~$ cd /var/www/html
sudo nano index.html
```



Step 8: View the website

- Refresh the browser



Conclusion

This experiment demonstrates how cloud platforms like AWS EC2 can be used to host web applications efficiently and make them publicly accessible.